

HOLD PRESSURE CURVE STUDY IN PLASTIC INJECTION MOLDING

Technical reference for engineers, processors, technicians, toolmakers, quality engineers,
and production leadership

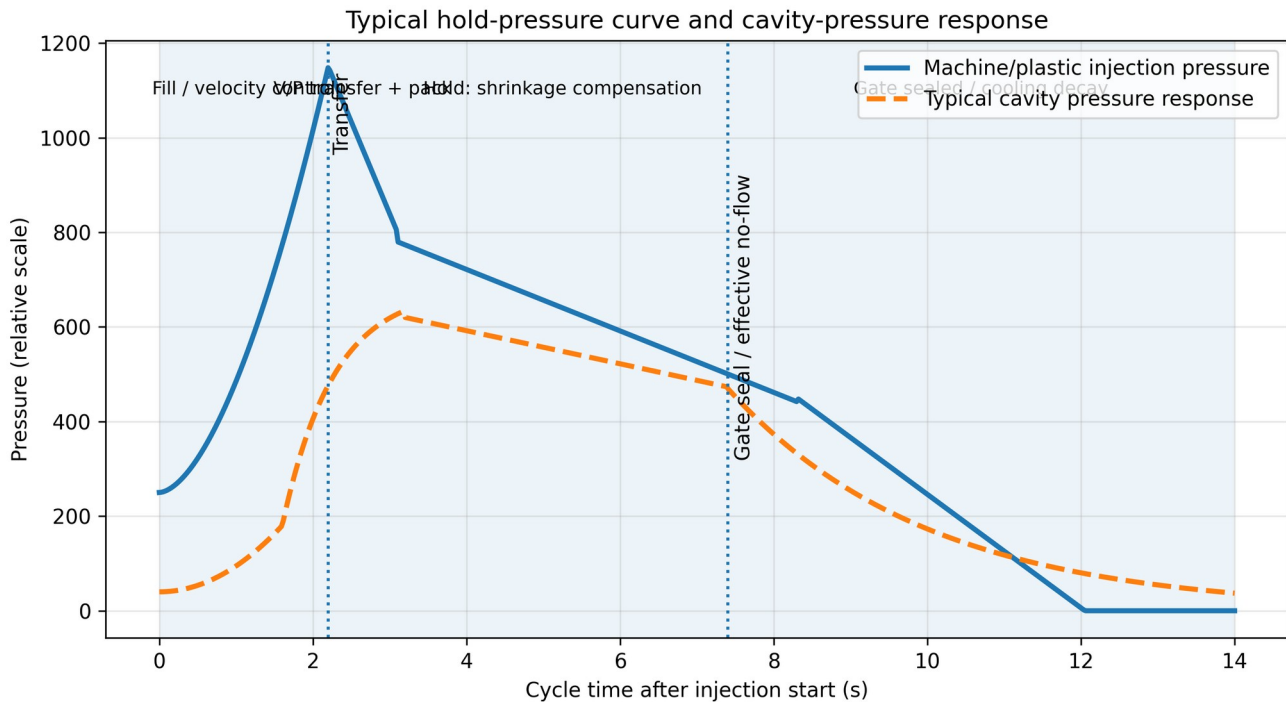


Figure 1. Original technical diagram: hold-pressure curve and cavity-pressure response over the molding cycle.

Prepared as an engineering reference from cited scientific-molding, machine-control, simulation, sensor, material, and standards sources. This report is ready for SME review, but it has not been externally peer-reviewed inside this chat.

Executive Summary

A hold pressure curve study is a controlled process-development experiment used to determine how second-stage pressure and time affect part weight, cavity pressure, shrinkage, sink, voids, flash, molded-in stress, dimensional capability, gate vestige, cycle time, and process robustness. In machine terms, it evaluates the pressure profile after velocity-to-pressure transfer. In part-quality terms, it evaluates the ability of the process to compensate polymer volumetric contraction while the gate or melt-entry system still allows pressure transmission. [1]-[8]

The most common shop-floor misunderstanding is treating the curve as a universal gate-freeze test. That is valid only for many cold-runner molds where the gate physically freezes and part weight reaches a clear plateau. Hot-runner and valve-gated molds behave differently. A hot tip or manifold stays molten, so a traditional part-weight plateau may never occur or may be misleading. Valve gates create a mechanical shutoff; the relevant variable is not just thermal freeze but the relationship among fill, pack, valve close timing, cavity pressure decay, and dimensional response. [4], [9]

A correct hold pressure curve does not produce a single magic pressure. It produces a defensible operating window. The target should be near the center of the window where dimensions, cosmetics, cavity-pressure signature, part weight, ejection, and cycle time are robust against normal variation in material viscosity, melt temperature, mold temperature, check-ring performance, water flow, and hot-runner thermal balance. [5], [10]-[14]

Safety and asset protection matter. Increasing pack/hold pressure can flash the mold, damage parting lines, collapse cores, stress slides, overload ejectors, create clamp-force deficits, and increase hot-resin release risk in hot-runner systems. The study must be run within validated machine, mold, hot runner, clamp, pressure, and material limits. Never defeat guards, safety interlocks, purge shields, hot-runner protective procedures, or documented mold-protection settings to run the experiment.

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1. Definition and Overview

Precise definition: A hold pressure curve is the plotted relationship between second-stage pressure and time after velocity-to-pressure transfer during injection molding, normally evaluated together with screw position, cushion, cavity pressure, part weight, and dimensional/cosmetic response. The study can be run as a hold-time sweep at fixed hold pressure, a hold-pressure sweep at fixed hold time, or a two-factor hold pressure-by-hold time design of experiments. Its purpose is to define the minimum and robust pressure/time profile that compensates resin shrinkage without overpacking or creating residual stress.

In a decoupled process, first-stage fill should fill the cavity primarily by velocity control, commonly to about 95-98% of intended fill depending on geometry, balance, and material behavior. Second stage then packs and holds the cavity while the polymer cools, densifies, and shrinks. Holding pressure is not meant to finish a short fill caused by bad transfer. It is intended to add controlled density and prevent backflow while the cavity is still pressure-coupled to the injection unit. [12]-[15]

Three curves are often discussed under the same umbrella. First is the machine actual pressure curve: hydraulic pressure or calculated plastic pressure versus time. Second is the cavity pressure curve: actual in-mold pressure versus time measured by a sensor. Third is the part response curve: part-only weight, dimensions, sink depth, void rate, or quality response versus hold pressure or hold time. A production-grade study should capture at least the machine curve and part response; high-cavitation, medical, optical, dimensional, and high-risk jobs should strongly consider cavity-pressure monitoring. [10], [11], [21], [22]

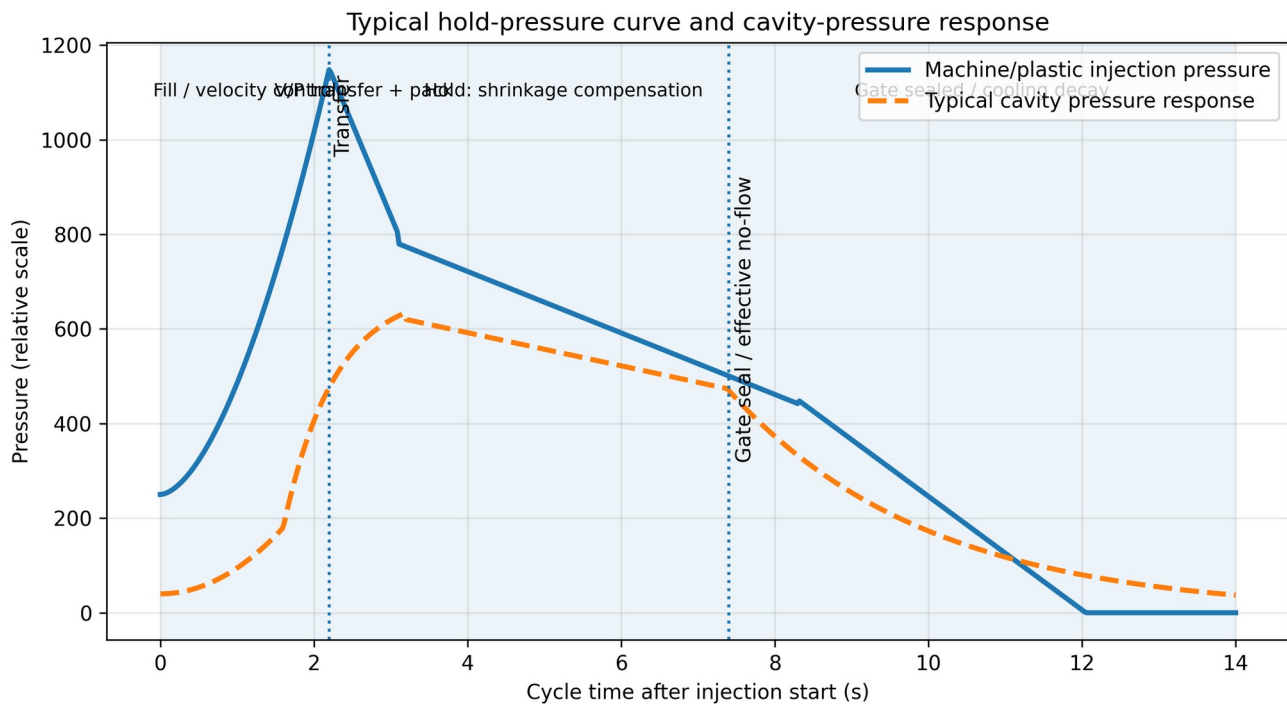


Figure 2. Machine-pressure and cavity-pressure curves are related but not identical. The part only sees the pressure transmitted through the melt path, gate, and freezing polymer.

A correct curve identifies three critical transition zones: the V/P transfer, where the press changes from velocity-dominant to pressure-dominant control; the effective packing window, where additional material or pressure can still influence the part; and the no-flow or no-useful-pressure-transmission point, where more hold time no longer improves the relevant part response. In cold-runner molds, this no-flow point often aligns with gate freeze. In hot-

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runner systems it may be a thermal, mechanical, rheological, or control-system limitation rather than a simple freeze event. [4], [6]-[9]

Term	Practical meaning	Measured by	Common mistake
Fill pressure	Pressure required to meet programmed fill velocity before transfer. It is a result of fill speed, viscosity, geometry, temperature, and restrictions.	Machine peak pressure, pressure at transfer, fill time, screw position.	Using pressure limit as if it were a velocity setting; masking short shots by overpacking.
Pack pressure	Higher-energy second-stage pressure immediately after transfer to finish volumetric filling and replicate detail.	Machine actual pressure; cavity pressure rise after transfer; part weight growth.	Packing with speed too low to reach pressure, creating a fake low-pressure study.
Hold pressure	Pressure maintained after pack to compensate shrinkage and prevent backflow until gate/no-flow point.	Pressure curve, screw movement, cavity-pressure decay, part weight/dimensions.	Holding long after no-flow point, wasting cycle time.
Gate seal / gate freeze	Condition where the cavity is no longer meaningfully pressure-fed through the gate.	Cold runner: part-weight plateau; sensors; simulation; short-shot/sectioning. Hot runner: more complex.	Assuming every mold produces a clean plateau.
Cavity pressure integral	Area under the cavity-pressure curve; often correlates with weight, dimensions, and molded-in density.	In-mold pressure sensors and monitoring system.	Ignoring curve shape and only watching peak pressure.

2. Technical Reasons for Running a Hold Pressure Curve

The phrase "root cause of why the study is done" should be treated as a production trigger. A hold pressure curve is run when the organization needs evidence, not guesswork, about whether the pack/hold portion of the process is capable. Typical triggers include dimensional drift, sink, voids, flash, gate blush, inconsistent weight, high scrap at startup, press-to-press transfer, resin lot changes, regrind changes, hot-runner balance issues, or uncertainty about whether cycle time is inflated by excessive hold. [1]-[5]

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Why a hold-pressure curve is run: common production drivers

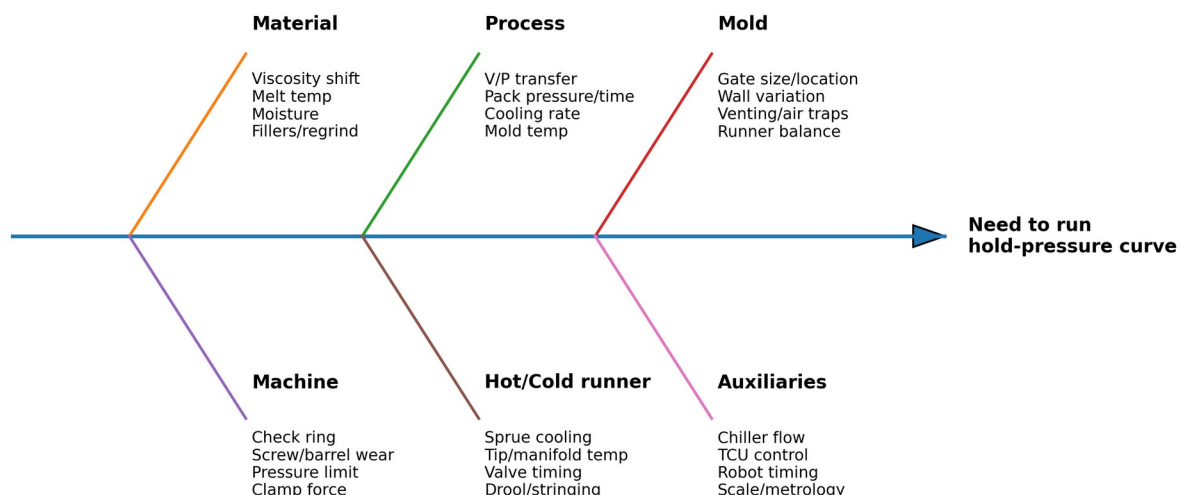


Figure 3. Cause map: production drivers that justify a hold-pressure curve study.

Production symptom	Why hold-pressure study is relevant	What the curve should reveal
Sink marks at ribs, bosses, thick sections	The local core shrinks after the skin freezes. More or better-timed pack may compensate if the gate is still open.	Whether weight/dimension/sink improves with pressure/time or whether gate/wall/cooling design prevents compensation.
Internal voids	Void forms when shrinkage cannot be fed and internal tensile stress exceeds melt strength during cooling.	Whether additional hold pressure/time removes voids or whether geometry/gate/cooling redesign is required.
Dimensional undersize or variable dimensions	Holding pressure controls density and shrinkage gradient.	Pressure/time zone where dimensions become capable without flash or stress.
Flash at parting line or vents	Pack pressure may exceed clamp/mold sealing capability.	Upper hold-pressure boundary and need for clamp, vent, shutoff, or mold repair action.
Brittleness, cracking, stress whitening, optical birefringence	Overpacking and high pressure gradients can increase molded-in stress.	Maximum safe pressure/time and need to trade shrinkage control against residual stress.
Cycle-time reduction project	Hold may be longer than useful no-flow time.	Shortest hold time that maintains acceptable part response with total cycle adjusted correctly.
Press transfer or new machine qualification	Hydraulic pressure numbers do not transfer directly across machines. Plastic pressure, fill time, and cavity-pressure fingerprint are better.	Machine-independent pressure response and new setpoint conversion.
Hot runner conversion or repair	Tips, drops, manifolds, and valve pins alter melt temperature and pressure balance.	Whether each cavity reaches similar pressure/weight/dimension and whether zones or valve timing need tuning.

3. Material, Process, Mold, Machine, and Auxiliary Factors

3.1 Material factors

Material viscosity determines how effectively hold pressure transmits through the runner, gate, and partially frozen melt. Higher viscosity, lower melt temperature, excessive moisture, degraded material, excessive fillers, or cold-runner restrictions reduce pressure transmission. Lower viscosity can improve pressure transmission but can also increase flash risk, gate blush, molecular orientation, and overpacking sensitivity. A hold pressure curve must therefore be tied to the actual resin grade, lot, colorant, regrind percentage, moisture condition, and melt temperature at the time of the study.

Melt temperature affects both flow and density. Raising melt temperature usually lowers viscosity and improves pressure transmission, but it also increases the amount of thermal contraction that must be compensated and may delay solidification. For semi-crystalline materials, higher melt and mold temperatures can change crystallization and shrinkage. For amorphous materials, the response is more dominated by thermal contraction, glass transition behavior, stress relaxation, and pressure history. [16]-[18], [28], [29]

Polymer family	How it typically changes the curve	Common shop-floor implication	Cold runner emphasis	Hot runner emphasis
PP / PE polyolefins	Semi-crystalline; relatively high shrinkage; gate-seal curve can be soft or slow to flatten, especially with large gates or soft olefins. Fillers and nucleation change shrinkage and freeze behavior.	Needs strong shrinkage compensation but can overpack near-gate areas. Weight plateau may be less crisp than in amorphous materials.	Separate part-only weight from runner weight. Watch runner cooling, sprue pull, and regrind contamination.	Tip temperature and manifold balance strongly affect pressure transmission; thermal gates may keep feeding enough to obscure a plateau.
ABS	Amorphous; moderate shrink; usually clearer weight response than PP. Sensitive to gloss, blush, weld/flow marks, and venting.	Can use hold pressure to improve gloss/density, but excessive pressure may show gate blush or molded-in stress.	Gate freeze may be measurable; monitor surface gloss and weld-line strength.	Watch residence time, hot spots, color streaks, and hot-runner shear heating.
PC	Amorphous engineering polymer; higher viscosity; lower shrinkage than PP but more stress-sensitive. Requires proper drying and elevated mold temperature for stress control.	Do not chase shrinkage with excessive hold pressure if the result is stress cracking, birefringence, or difficult ejection.	Large sprue/runner may control cycle. Gate size must support pressure without excessive shear.	Hot-runner thermal uniformity, degradation avoidance, and valve-gate timing are critical; cavity-pressure sensors are valuable.
Glass-filled nylon / PBT / PET	Anisotropic shrinkage, fiber orientation, moisture/drying sensitivity, and crystallinity can dominate dimensional response.	The curve may improve weight while worsening warp if fiber orientation and cooling are poor.	Gate freeze and runner freeze must be evaluated with conditioned dimensions.	Hot-runner shear and thermal history can affect fiber length/orientation and cavity balance.
TPE / TPU / soft materials	Compressible, elastic, and sometimes slow to stabilize; pressure response can be broad rather than sharp.	Part weight plateau may be hard to interpret; dimensions may relax after ejection.	Use longer conditioning and good weighing protocol.	Tip drool, stringing, and valve close timing can dominate results.

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Fillers and additives change both the material response and the measurement interpretation. Glass fiber raises viscosity and creates anisotropic shrinkage. Mineral filler often lowers shrinkage but raises abrasion risk. Lubricants and flow promoters may improve fill but can change weld-line strength and plate-out. Flame retardants and colorants can alter degradation sensitivity. Regrind can lower molecular weight and shift viscosity; if regrind percentage varies during a study, the curve may document material variation rather than the true hold-pressure window.

Moisture is a hidden confounder. Hygroscopic resins such as PC, ABS blends, nylon, PBT, PET, TPU, and many medical grades require validated drying. Hydrolysis or splay from wet resin can cause flow, strength, surface, and dimensional changes that look like pack/hold defects. Do not run a hold pressure curve until dryer dew point, drying time, hopper residence, and material handling are under control.

3.2 Process parameters

Injection speed and transfer position control how much material is in the cavity before hold begins. A hold pressure study is invalid if first-stage fill is unstable, pressure-limited, short, overfilled, or transferred by time only when screw recovery and viscosity vary. Start with a stable fill-only process and a defensible V/P transfer. Any change in fill speed or transfer position changes the baseline mass, shear heat, frozen layer thickness, and cavity pressure at the beginning of hold.

Cooling rate controls how long the gate, wall, and thick features remain pressure-transmissive. Higher mold temperature can delay freeze and improve surface/stress performance, but it can also increase cycle time and shrinkage sensitivity. Lower mold temperature can freeze the gate early and make hold pressure ineffective before the thick section is packed. A hold pressure curve must record actual mold-water supply/return temperatures and, preferably, flow rate and cavity steel temperature.

Parameter	Too low / too short	Too high / too long	Best-practice study rule
Hold pressure	Sink, voids, low weight, undersize dimensions, backflow through gate.	Flash, overpacked near-gate dimensions, high stress, ejector marks, mold damage.	Sweep pressure after fill is stable; identify lower and upper quality boundaries.
Hold time	Backflow, weight drift, sinks, variable dimensions, unstable cavity pressure.	Cycle waste, residual stress, gate blush, no quality improvement after no-flow point.	For cold runners, continue until weight plateau; for hot runners, use pressure/dimensional window.
Second-stage speed / pressure response	Machine may be speed-limited and never reaches set pressure, creating false low-pressure result.	Can hammer the cavity after transfer if transition is too abrupt.	Verify actual pressure reaches setpoint promptly without overshoot outside tool limits. [13]
Melt temperature	High viscosity, short pressure transmission, early gate freeze, poor surface.	Degradation, longer freeze, greater shrinkage to compensate, flash risk.	Use resin supplier range and measure actual melt temperature, not only barrel setpoints.
Mold temperature	Early freeze, sink not correctable, stress in amorphous materials, poor surface replication.	Longer cycle, larger shrinkage in crystalline materials, possible sticking.	Run curve only after water circuits and steel temperature stabilize.
Cooling time	Parts too soft; dimensions move after ejection; ejection distortion.	Wasted cycle; masks inadequate hold study because part cools long enough despite poor pack.	When changing hold time, adjust cooling time to maintain total cycle if the goal is gate-seal isolation.

3.3 Mold design factors

Gate design is the strongest mold-design lever for the study. A thin gate freezes early and limits how long hold pressure can influence the part. A large gate extends the packing window but can create high gate vestige, stress, and long cycle time. Literature on gate freeze-off shows gate thickness is a major determinant of sealing time, and cavity geometry also matters. [6]

Gate location controls pressure gradient and shrinkage distribution. Gating into a thick section allows more effective packing; gating into a thin section that feeds a thick boss or rib can make sink almost impossible to eliminate with process settings. Multiple gates can create uneven packing, weld lines, unbalanced pressure decay, and local overpack/underpack zones.

Wall thickness variation converts a simple pressure study into a geometry problem. Uniform walls produce more predictable pressure decay. Thick-to-thin-to-thick flow paths cause early freeze in the thin region, isolating the thick region from hold pressure. Ribs, bosses, handles, hinge barrels, child-resistant closure features, and deep collars are common zones where the curve reveals a design limitation rather than a process setting error.

Venting and air traps matter because trapped gas compresses during fill and pack. Insufficient venting can show burns, dieseling, short shots, glossy/dull shifts, weak weld lines, and pressure spikes. Increasing hold pressure to overcome trapped air is a poor correction; it can flash vents, overpack other areas, and hide the real issue until material lot or ambient conditions change.

3.4 Machine factors

Machine feedback must be trusted before the curve is trusted. Screw and check-ring leakage can produce falling actual pressure, cushion drift, shot-to-shot weight variation, and inconsistent response at the same setpoint. Barrel wear, poor screw design, worn flights, excessive back pressure, and unstable recovery can change melt temperature and homogeneity. A hold pressure curve run on a leaking non-return valve may lead to a false conclusion that the mold needs more pressure.

Plastic pressure should be distinguished from hydraulic pressure. Hydraulic pressure is machine-cylinder pressure. Plastic pressure is the pressure applied to the melt and depends on machine geometry and intensification ratio. A setting of 500 psi hydraulic on one press may not equal the same plastic pressure on another press. For process transfer, document plastic pressure, screw position, fill time, cushion, and actual pressure curve where available. [12]

Clamp force must exceed the opening force created by projected area multiplied by cavity pressure, with a suitable safety factor and tool-specific considerations. A hold pressure sweep that approaches clamp limits will find flash before it finds the true upper packing boundary. Flash created during a study can damage vents and parting lines. If flash appears only at high pressure, the result is still valuable: it defines the upper boundary of the process window.

Machine item	Failure mode during study	How to detect	Required action
Check ring / non-return valve	Melt leaks backward during hold; cushion drifts; part weight scatter high.	Cushion variation, pressure decay, check-ring repeatability test, short-shot repeatability.	Repair or compensate only after risk assessment; do not validate a process on a bad valve.
Screw/barrel wear	Inconsistent melt temperature, unmelt, long recovery, poor color mixing.	Recovery trend, melt temperature shot-to-shot, black specks, screw pull inspection.	Schedule maintenance; avoid interpreting material/mold from bad melt prep.
Pressure limit / injection unit	Actual hold pressure cannot	Actual pressure trace flat at	Use larger press/injection

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Machine item	Failure mode during study	How to detect	Required action
size	reach setpoint or transfer is pressure-limited.	limit; screw stalls; alarm history.	unit or redesign gate/runner/material/process .
Clamp force	Flash or mold opening at high hold pressure.	Tie-bar strain, flash at parting line, projected-area calculation, clamp tonnage trend.	Reduce pressure, repair mold, adjust clamp, or use proper tonnage.
Controller response	Overshoot/undershoot during V/P transfer; inconsistent pack.	Pressure curve shape, controller logs, pressure transition time.	Tune transfer, second-stage speed, pressure ramp, or machine control mode.

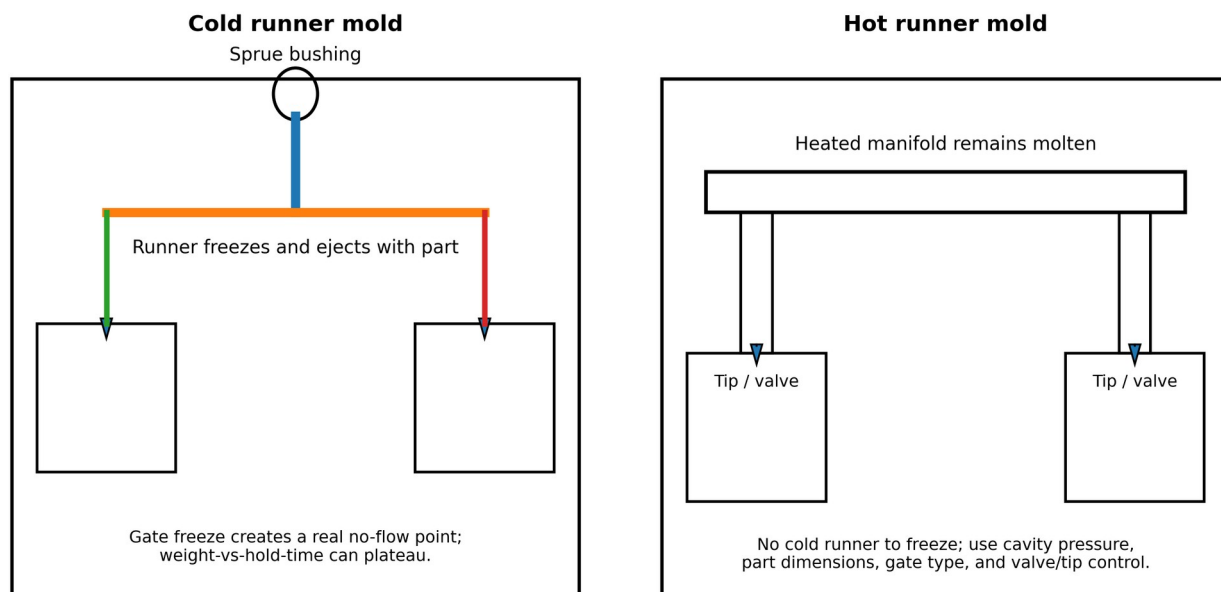
3.5 Auxiliary equipment factors

Chillers and thermolators influence the study by changing actual steel temperature, not merely water setpoint. A thermolator with poor flow or air in the circuit may show correct supply temperature but fail to remove heat. A chiller that short-cycles or runs near capacity can shift mold temperature during the curve. Record supply temperature, return temperature, flow rate, and pressure drop for critical circuits.

Hot-runner temperature controllers are not passive accessories. They define melt temperature at every drop, manifold branch, sprue bar, and tip. A one-zone drift can create a cavity-specific hold-pressure response that looks like mold imbalance. Thermocouple mislocation, heater failure, poor PID tuning, wrong boost/standby settings, and leaking valve-pin seals can all corrupt the curve. Hot-runner faults are a major reason hot-runner studies should use cavity-by-cavity data instead of total shot weight alone. [9]

Robots, sprue pickers, conveyors, degaters, and take-out systems can distort a study if they change cooling, gate vestige, part handling, or conditioning. A sprue picker delay can extend mold-open time and change the next shot's steel temperature. A robot that grips a soft part can induce warp. Cold-runner molds are especially sensitive to sprue/runner handling, while hot-runner molds are sensitive to part take-out consistency because there is no runner mass to buffer thermal changes.

4. Cold Runner vs Hot Runner Interpretation



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Figure 4. Cold-runner and hot-runner melt-delivery systems create different hold-pressure study physics.

Cold runner molds have a solidifying sprue and runner network. Because the gate and runner cool with the part, the part can become isolated from the machine once the gate freezes. This makes a part-weight versus hold-time plateau a useful method in many cold-runner tools. The operator increases hold time at fixed, acceptable hold pressure, weighs parts only after steady-state shots, and identifies the earliest hold time after which part-only weight no longer increases meaningfully. The production setpoint is commonly placed slightly above that value, subject to dimensional, cosmetic, and cycle-time validation. [2]-[4], [15]

In a cold-runner study, never let runner weight hide part response. If the runner is weighed with the part, a thick sprue or runner can dominate the weight curve. A sprue can continue to densify or cool even when the part gate is effectively sealed. For multi-cavity molds, individual cavity weights are more valuable than total shot weight. A balanced total shot can still hide one underpacked cavity and one overpacked cavity.

Hot runner molds keep the delivery system molten. The manifold and drops maintain melt temperature; there is no cold runner to freeze and eject. Husky describes hot runners as a heated delivery network with manifolds and nozzles that maintain molten plastic, reducing waste and cycle time while adding thermal and control complexity. [9] In this context, a conventional gate-seal weight plateau is often invalid. FimmTech explicitly distinguishes cold-runner plateau behavior from hot-runner or valve-gated systems where the gate area remains molten and the part-weight curve may not flatten. [4]

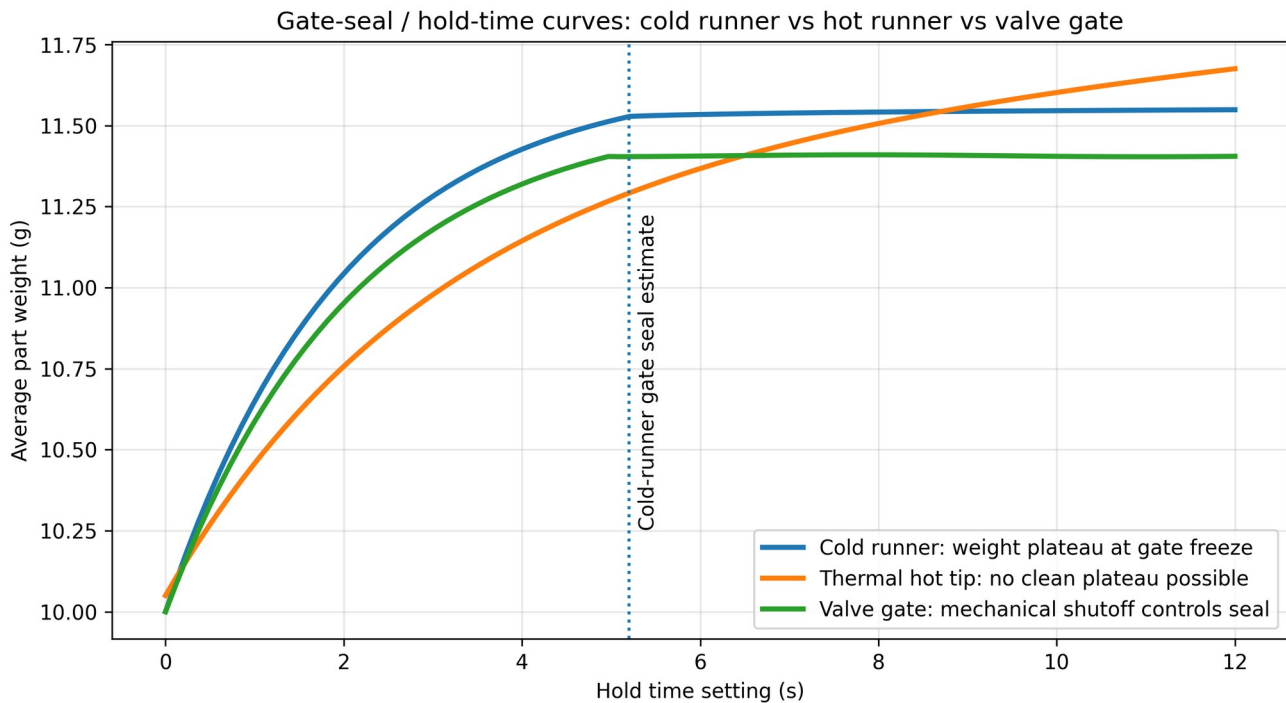


Figure 5. The same part-weight-versus-hold-time logic is not equally valid for all melt-delivery systems.

Thermal hot tips require a quality-window approach. Hold pressure and time should be evaluated against part-only weight, dimensions, gate vestige, sink, flash, cavity-pressure curves, tip temperature, manifold balance, and thermal drool/stringing. A shallow continuous weight increase at longer hold times may indicate ongoing pressure transmission through a hot tip rather than useful quality improvement. The selected setting should be a robust center point, not the longest time tested.

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Valve-gated molds create a mechanical seal. The valve close event can end cavity feed even if the melt remains hot. The curve must therefore include valve open time, valve close timing, hydraulic/pneumatic/servo actuation delay, pin wear, and cavity-pressure behavior before and after closure. If the valve closes too early, the part underpacks; if it closes late under high pressure, the part can overpack or show gate defects. Multi-drop sequential valve-gated molds require gate-by-gate timing studies, not just global hold-pressure settings.

Study issue	Cold runner mold	Hot tip mold	Valve-gated hot runner mold
Gate seal indicator	Often part-only weight plateau vs hold time.	No reliable plateau; use cavity pressure, dimensions, sink, and process window.	Valve close creates mechanical no-flow; verify pressure/dimension response before and after close.
Cycle driver	Runner/sprue cooling may drive cycle.	Part cooling and hot-runner thermal stability drive cycle.	Part cooling, valve timing, actuator response, and vestige control drive cycle.
False conclusion risk	Runner weight mistaken for part weight; sprue freeze mistaken for gate seal.	Continuous hot-tip feed mistaken for useful packing.	Valve timing mistaken for thermal freeze.
Main auxiliary risk	Sprue picker, degater, regrind handling.	Hot-runner controller, thermocouple, heater, leakage, tip balance.	Controller plus valve-pin actuation, pin wear, sequencing, gate damage.
Best measurement	Part-only cavity-by-cavity weight, dimensions, visual, machine pressure.	Cavity pressure, part-only weight, dimensions, hot-runner zone data.	Cavity pressure at each critical cavity, valve timing trace, dimensions, vestige.

5. Real-World Examples and Case Studies

This section separates verified technical cases from informal practitioner accounts. Informal public forum accounts can be useful for training because they show how molders think through real problems, but they are not peer-reviewed evidence. They should not override controlled data, supplier guidance, or plant validation.

5.1 Medical device transfer: scientific molding process transfer

A peer-reviewed medical-device case study by Fitzgerald, McDonald, Devine, and Fuenmayor documented the transfer and optimization of an injection molding process from an Arburg 221M to an Arburg 375 V for an endovascular aneurysm repair dilator device, where a polypropylene hub was molded over a high-density polyethylene insert. The authors described scientific molding as a method for developing robust processes resilient to viscosity and temperature variation and emphasized collection of fill, pack/hold, temperature, and pressure data. Their transfer work included material rheology, gate freeze, cavity balance, pressure loss, and DOE activities. [5]

Production lesson: for regulated medical work, a hold pressure curve is not a standalone tribal-knowledge exercise. It becomes part of process characterization, transfer justification, dimensional capability, and documented validation. The curve should support a manufacturing control strategy: what is monitored, what alarms, what gets quarantined, and what process changes trigger revalidation.

5.2 Hot-runner / valve-gated production: why the classic plateau fails

FimmTech provides a practical hot-runner example: in cold runners, a weight plateau can identify hold time, but in hot-runner or valve-gated systems the gate area remains molten and the part-weight curve may not flatten. The recommended approach shifts to selecting a dimensional process window rather than applying a cold-runner gate-freeze rule blindly. [4]

Production lesson: if a processor expects a plateau on a hot-tip or valve-gated system, they may keep increasing hold time and pressure until flash, stress, gate vestige, or cycle waste appears. For hot runners, pressure and temperature control by drop, plus cavity-specific data, provide a more reliable interpretation than total shot weight.

5.3 Simulation-supported packing-time decision

Moldex3D documents a method to estimate gate freeze time by using criteria such as gate-section temperature, gate flow rate, and part weight change rate, and it reports that insufficient packing can lead to excessive cooling-volume shrinkage, uneven shrinkage, warpage, and sink marks. Its Pack capability includes predicting gate freeze time, packing time, packing pressure, clamp-force requirement in packing, and regions of high volumetric shrinkage. [7], [8]

Production lesson: simulation should not replace a press-side curve, but it can prevent bad trial plans. Before steel is cut or before a risky mold trial, simulation can identify gates that freeze too early, thick regions that cannot be packed, high volumetric shrinkage zones, and clamp-force risk at proposed hold pressures.

5.4 Automotive/optical stress case logic

Optical and automotive lens parts illustrate the other side of hold pressure: more packing is not always better. Supplier guidance for minimizing molded-in stresses in optical polycarbonate-type parts typically includes reducing packing pressure and packing time when stress fringes or cracking risk appear, while also using proper mold and melt temperatures. [16]

Production lesson: an automotive bezel, lens, or display cover may pass weight and dimensional checks while failing stress, birefringence, coating, or environmental exposure requirements. The hold pressure curve must include the response that matters to the product, not just the response that is easy to measure at the press.

5.5 Public practitioner accounts

A public r/InjectionMolding discussion on gate freeze-off described a molder experimenting with low hold pressure, ABS behavior, and a hot sprue feeding a cold sub-runner/cavity layout. The discussion is not an audited case study, but it reflects a frequent real-world issue: hybrid runner systems can behave neither like a pure cold runner nor like a fully controlled hot-runner valve-gated system. [30]

A public Eng-Tips discussion about nondestructive void inspection included the practical observation that void avoidance is usually better than trying to inspect every molded part, and that constant thickness plus proper packing under time and pressure are central. Again, this is informal, but it aligns with standard processing logic: do not use NDT as a substitute for a capable pack/hold process. [31]

6. Diagnostic Techniques

6.1 How to tell whether a hold pressure curve has actually been run

A physical part usually cannot prove that a hold pressure curve was run. Good parts can be made by accident, and bad parts can come from a previously validated process that has drifted. The strongest evidence is a data package: dated study sheet, material lot and dryer data, machine ID, mold ID, cavity map, transfer position, fill time, cushion, actual pressure curves, hold-pressure/time table, part-only weights by cavity, dimensional results after conditioning, visual defect notes, and final process window rationale.

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Visual inspection can suggest whether pack/hold is wrong. Underpacked parts may show sink, voids, low gloss over thick sections, short-shot vestiges, and variable dimensions. Overpacked parts may show flash, gate blush, ejector pin witness marks, stress whitening, stuck parts, gate protrusion, or excessive near-gate dimensions. However, visual inspection is qualitative; it should trigger data review, not replace the curve.

Microscopy, cross-sectioning, and destructive checks can reveal internal voids, weld-line weakness, poor fiber distribution, and frozen-layer effects. For PC and transparent amorphous resins, polarized-light inspection can reveal stress patterns. For filled materials, sectioning and microscopy can show orientation gradients and packing-related density differences. These methods are useful for validation and failure analysis, but they are slower than in-process pressure and weight monitoring.

6.2 Moldflow / Moldex3D / CAE simulation

Moldflow, Moldex3D, Cadmould, Sigmasoft, and similar software can simulate filling, packing, cooling, volumetric shrinkage, pressure distribution, clamp force, gate freeze, air traps, weld lines, fiber orientation, and warpage. Simulation output must be interpreted with real material data, realistic hot-runner thermal assumptions, cooling-channel conditions, venting assumptions, and accurate geometry. Moldex3D specifically documents packing analysis functions for gate freeze and packing optimization. [7], [8]

The user prompt mentioned simulating a viscosity curve. A viscosity curve and a hold pressure curve are related but different. A viscosity curve normally studies fill speed versus relative viscosity/pressure response during fill to select a robust fill rate. A hold pressure curve studies second-stage pressure/time response after transfer. The fill curve should usually be completed before the hold curve; otherwise the hold study may compensate for a poor fill rate.

6.3 Alternatives that can produce equal or better evidence

Method	What it measures	Strength	Weakness
Part-weight hold-time curve	Average part-only weight versus hold time.	Simple and cheap; good for many cold-runner tools.	Misleading for hot runners, valve gates, multi-cavity imbalance, and thick runners.
Hold-pressure sweep	Part response versus hold pressure at fixed time.	Finds lower/upper pressure boundaries for sink, flash, stress, and dimensions.	Requires stable transfer and time setting; can damage molds if limits ignored.
Two-factor DOE	Hold pressure and time together, possibly with mold temperature or transfer.	Maps robust window and interactions.	More parts, measurements, and analysis required.
Cavity-pressure sensors	Actual pressure in the cavity versus time.	Best direct fingerprint for fill/pack/hold quality and automated alarms.	Requires sensorized mold, installation discipline, and interpretation.
Screw-position/cushion analysis	Machine-side movement during hold.	Useful for check-ring leakage and pressure-limited detection.	Indirect; not a substitute for cavity data.
Closed-loop cavity-pressure control	Controls hold/pack to a pressure curve in the cavity.	Can compensate viscosity variation and machine drift.	Requires sensors, controller capability, and validation.
Bayesian/adaptive DOE	Uses prior runs to select next best experiment.	Can reduce experimental burden for complex windows.	Requires data infrastructure and statistical competence.

6.4 Non-destructive testing for internal defects

NDT can assess consequences of bad packing, but it usually cannot replace process characterization. Radiographic or industrial CT inspection can reveal voids, inclusions, short packs, and internal geometry. Ultrasonic testing and scanning acoustic microscopy can detect voids, delamination, cracks, and weld-line features in suitable materials

Prepared for technical training and process development; validate against plant-specific mold, machine, material, and safety requirements.

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and geometries. General NDT bodies list radiographic and ultrasonic testing as major methods for internal defect detection, and research literature has applied ultrasound and scanning acoustic microscopy to injection-molded components and process monitoring. [23]-[25]

NDT / inspection method	Best use	Limitations for hold-pressure diagnosis
Industrial CT / X-ray	Internal voids, thick-section shrinkage, dimensional metrology, hidden short shots.	Costly; sampling-based; cannot show pressure history by itself.
Ultrasonic testing	Internal discontinuities, wall thickness, some crystallinity/pressure-related monitoring research.	Geometry, coupling, and material attenuation can limit resolution.
Scanning acoustic microscopy	Fine voids, delamination, cracks, weld-line inspection on small/flat components.	Lab method; slow for production; requires skilled interpretation.
Polarized light	Residual stress in transparent amorphous materials such as PC.	Not applicable to opaque materials; qualitative unless calibrated.
Weight and density	Bulk packing level and cavity balance.	Cannot locate voids; hot-runner/multicavity data can mislead if aggregated.
Sectioning / microscopy	Definitive for local voids, weld lines, fiber orientation, gate freeze evidence.	Destructive; not NDT; still valuable for validation.

7. Preventive Measures and Optimization Actions

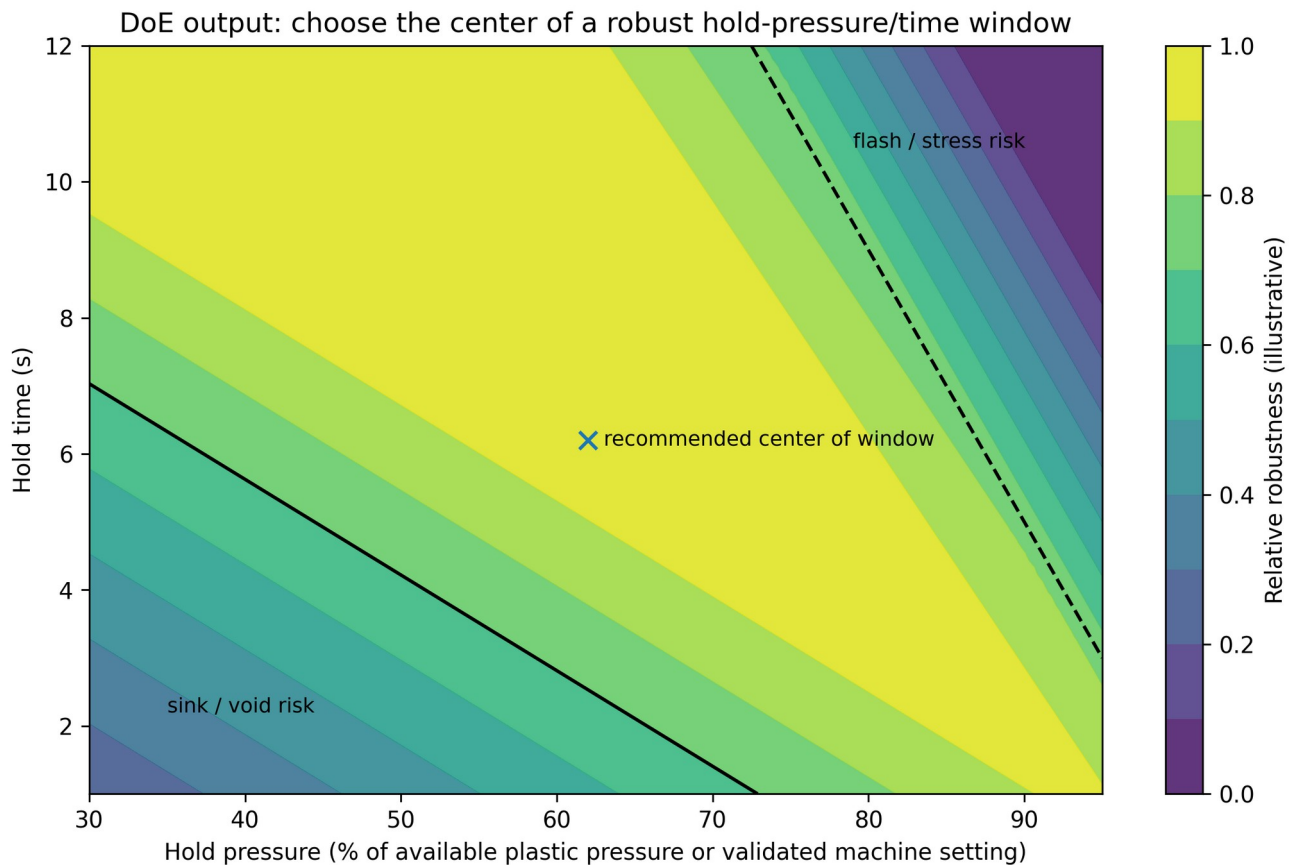
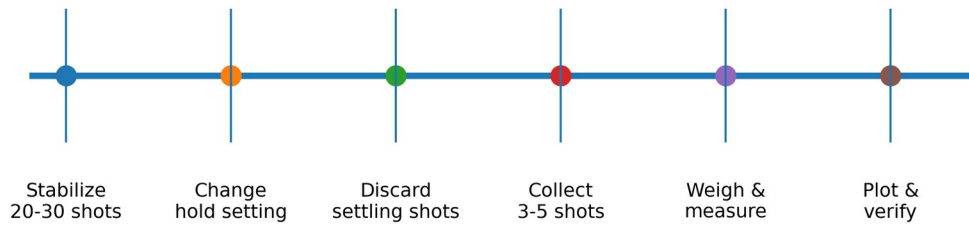


Figure 6. A pressure/time DOE should identify the center of a robust process window, not the highest pressure or shortest cycle by itself.

7.1 Before running the DOE

Prevention starts before the curve. The press, mold, material, and auxiliaries must be in steady state. A curve collected during warm-up is usually a record of thermal drift. Stabilize the mold and hot runner, verify water flow, dry the resin, purge thoroughly, use a known material lot, verify the scale, and run enough cycles for pressure, cushion, part temperature, and dimensions to settle. The minimum data discipline is shown below.

Minimum data discipline for a defensible hold-pressure curve



Do not mix cavities, runner weights, unstable temperatures, or non-conditioned dimensions unless the study plan explicitly calls for it.

Figure 7. Sampling discipline for defensible hold-pressure data.

Readiness item	Requirement before study	Reason
Material	Correct grade, lot, color, dryer condition, moisture spec, regrind percentage, and residence time documented.	Material variation shifts viscosity and shrinkage response.
Melt temperature	Actual melt temperature measured by agreed method, not just barrel setpoints.	Barrel setpoints can differ from real melt temperature.
Mold temperature	Supply/return temperature, flow, and steel temperature stabilized.	Gate freeze and shrinkage depend on cooling conditions.
Hot runner	All zones at setpoint; no alarms; valve pins actuate correctly; manifold/tip settings recorded.	Hot-runner thermal imbalance alters pressure response.
Machine	Cushion stable, check ring good, pressure capability adequate, fill-only process repeatable.	Bad machine repeatability invalidates the curve.
Measurement	Scale resolution adequate; gage R&R or measurement method acceptable; cavity map prepared.	Data scatter must be smaller than the effect being studied.
Safety	Clamp force, mold protection, pressure limits, purge shield, hot-runner procedures, and guarding verified.	Hold-pressure sweeps can damage tooling or injure personnel if limits are ignored.

7.2 Material selection

Select resin with flow characteristics that match gate size, wall thickness, flow length, and product requirements. A high-flow grade may reduce fill pressure but increase flash and reduce mechanical performance. A lower-flow grade may improve toughness but require higher pressure, bigger gates, hotter melt, or larger press capacity. Supplier processing guides often recommend pack/hold pressure as a percentage of injection pressure, but this must be treated as a starting point, not a validated setting. [17], [18]

Additives can help but should not be used as unmanaged process fixes. Lubricants and flow promoters can improve flow, but they may affect weld-line strength, plating/painting, medical compliance, and long-term properties. Nucleating agents can shorten cycle and alter shrinkage in PP. Glass or mineral fillers reduce some shrinkage but introduce orientation and tooling wear. Every material change should trigger at least a reduced hold-pressure verification study.

7.3 Process optimization after the curve

The curve tells the engineer how part response changes with hold pressure and time. Use that information to set the production profile at the middle of the acceptable window, then define upper and lower alarms. If underpacking defects disappear by 50% hold pressure and flash begins at 80%, do not run at 78% unless there is a validated reason. Center the process where normal viscosity, mold temperature, and check-ring variation do not cross a quality boundary.

Multi-stage hold can be useful. A common approach is higher initial pack pressure immediately after transfer to finish detail and compensate early shrinkage, followed by lower hold pressure to maintain density without overpacking the gate area. For parts with thick sections far from the gate, however, a high initial pressure may overpack near-gate areas before the far thick section is adequately fed. The correct profile must be based on actual response, ideally cavity pressure at near-gate and end-of-fill locations.

Cold runner adjustment logic: after a cold-runner gate-seal curve, set hold time just above the plateau point, then reduce cooling time by the same amount if the objective is total cycle control and if the part is stiff enough at ejection. Increase hold pressure only if sink/void/dimension data require it and if clamp/mold limits allow it. Verify runner and sprue cooling because the runner may still control cycle.

Hot runner adjustment logic: do not chase a non-existent plateau. Tune manifold/tip temperatures, valve timing, and cavity balance. Use part-only dimensions, cavity-pressure curves, weight by cavity, and gate quality. If one cavity is light, raising global hold pressure may overpack every other cavity. Correct the hot-runner drop, tip, valve, or balance issue first.

7.4 Mold design improvements

- Increase gate thickness or reduce land length if gate freezes before the thick section packs, subject to gate vestige and shear limits.
- Move the gate toward the thickest functional section or add sequential valve gating if long-flow geometry creates uncontrolled pressure gradients.
- Redesign thick bosses, ribs, and pads using coring, rib ratio control, and uniform wall principles; do not expect hold pressure to fix poor geometry indefinitely.
- Improve venting at end of fill, weld-line locations, ribs, and shutoffs so packing pressure is not used to compress trapped gas.
- Balance cold runners by geometry and rheology, not just equal length; for hot runners, validate thermal and pressure balance by cavity.

7.5 Machine calibration and maintenance

- Perform check-ring repeatability tests and trend cushion. A hold curve is not valid when the non-return valve leaks.
- Inspect screw/barrel condition when recovery time, unmelt, black specks, or melt-temperature variation increases.
- Calibrate injection pressure, screw position, temperature zones, transducers, load cells, and cavity-pressure sensors at defined intervals.
- Verify heater bands, thermocouples, and barrel-zone control. A bad zone can alter viscosity enough to shift the whole curve.
- Use plastic pressure or cavity pressure for transfer between machines; do not transfer hydraulic pressure numbers blindly.

7.6 In-process adjustments

Real-time tweaks should be limited by the validated window. Operators should not freely adjust hold pressure to chase every short-term defect unless the control plan defines the limits and reaction plan. A proper process sheet should include allowed adjustments, required approvals, sample size after adjustment, and whether the adjustment triggers quality hold or engineering notification.

For hot runners, real-time adjustments may include tip temperature trims, valve timing, or manifold zone balancing. These changes can affect material residence time, shear, color stability, drool, and cavity balance. Use a documented hot-runner change log. A hot-runner temperature change is a process change, not a harmless convenience.

7.7 Post-processing solutions

Post-processing can sometimes salvage cosmetics, but it should not be treated as a process-control substitute. Polishing, painting, plating, trimming, gate deflashing, annealing, or machining can hide symptoms while leaving residual stress, internal voids, weak weld lines, or dimensional instability. Use post-processing only after engineering confirms that functional requirements are still met.

Issue after study	Cold runner post-processing option	Hot runner post-processing option	Risk
Gate vestige	Trim, cut, tumble, local polish.	Valve gate vestige polish, pin maintenance, trim if permitted.	Can mask overpack or poor gate design.
Surface sink / blush	Paint or texture adjustment in non-critical cosmetics.	Tip temperature/vestige improvement before painting.	Paint may reveal sinks more strongly; structural void remains.
Flash	Deflash or trim for temporary containment.	Gate drool/string trim if isolated.	Deflashing hides mold damage; inspect shutoffs and clamp.
Residual stress	Anneal if material/product allows.	Anneal plus hot-runner thermal correction.	Annealing can change dimensions; must be validated.

7.8 Redesign strategies

When the curve shows no acceptable window, redesign is required. Typical redesign actions include increasing gate size, moving the gate, adding a second gate or valve sequence, coring thick sections, reducing rib-to-wall ratio, adding radii, improving venting, moving cooling channels closer to thick zones, changing resin, or converting cold runner to hot runner. A process engineer should not own a geometry problem alone; tooling, design, quality, material supplier, and production must participate.

Prepared for technical training and process development; validate against plant-specific mold, machine, material, and safety requirements.

8. Industry Best Practices

The strongest industry pattern is scientific molding: characterize material viscosity/fill behavior, cavity balance, pressure drop, cosmetic process window, and gate seal/hold behavior before locking a production process. FimmTech describes a six-step methodology including viscosity curve, cavity balance, pressure drop, cosmetic process window, gate seal, and process window. [1] Plastics Technology and Paulson Training likewise emphasize second-stage pack/hold optimization and gate-seal determination as part of robust process setup. [2], [3], [15]

Use standards where they apply. ISO 294-1 addresses reproducible injection molding of thermoplastic test specimens, and ASTM D955 addresses shrinkage measurement from mold dimensions to molded dimensions under specified process conditions. These standards do not give a universal hold-pressure setting for production parts, but they reinforce the principle that molding conditions must be documented and controlled if dimensions and properties are to be comparable. [19], [20]

Use manufacturer control technologies when the risk justifies them. ARBURG's ReferencePilot controls the holding-pressure curve in real time using a pressure sensor in the mold and is positioned to counter viscosity fluctuations and screw wear. [10] Kistler describes in-mold piezoelectric cavity-pressure sensors as a basis for pressure-curve measurement, process monitoring, and optimization. [11] RJG's decoupled process materials emphasize separating fill, pack, and hold and using pressure curves to understand what the plastic is doing. [12]-[14]

Quotes and interviews: This report does not invent interviews. Where practitioner voice is useful, the report uses public practitioner discussions and labels them as informal. For a plant training program, the best practice is to interview your own senior processors, tooling engineers, and quality engineers, then attach their signed review notes to this document. That is how the document becomes genuinely peer-reviewed for your plant, material base, molds, and machines.

9. Emerging Technologies and Research

Smart molding is moving the hold pressure curve from a one-time DOE into continuous control. In-mold cavity pressure, temperature sensors, tie-bar strain, screw position, injection power, melt temperature, hot-runner zone data, and quality results can be logged every cycle. The practical objective is to reject or correct a bad cycle before bad parts accumulate. Cavity pressure is particularly valuable because it measures the pressure history the part actually experienced. [10], [11], [21], [22]

AI and adaptive optimization can reduce the number of DOE points and improve process control, but only if the data are clean. Recent research includes Bayesian adaptive DOE for injection molding optimization and machine-learning closed-loop process control. These methods are promising for cycle time, shrinkage, warpage, and quality prediction, but they require structured data, validated sensors, change control, and a clear definition of acceptable quality. [26], [27]

Material-science innovations such as nano-additives, tailored fillers, recycled-content stabilizers, bio-based polymers, and self-healing concepts can change viscosity, crystallization, thermal conductivity, shrinkage, moisture sensitivity, and degradation behavior. The processing implication is simple: any new formulation invalidates old assumptions about hold pressure until the curve is rerun or technically bridged.

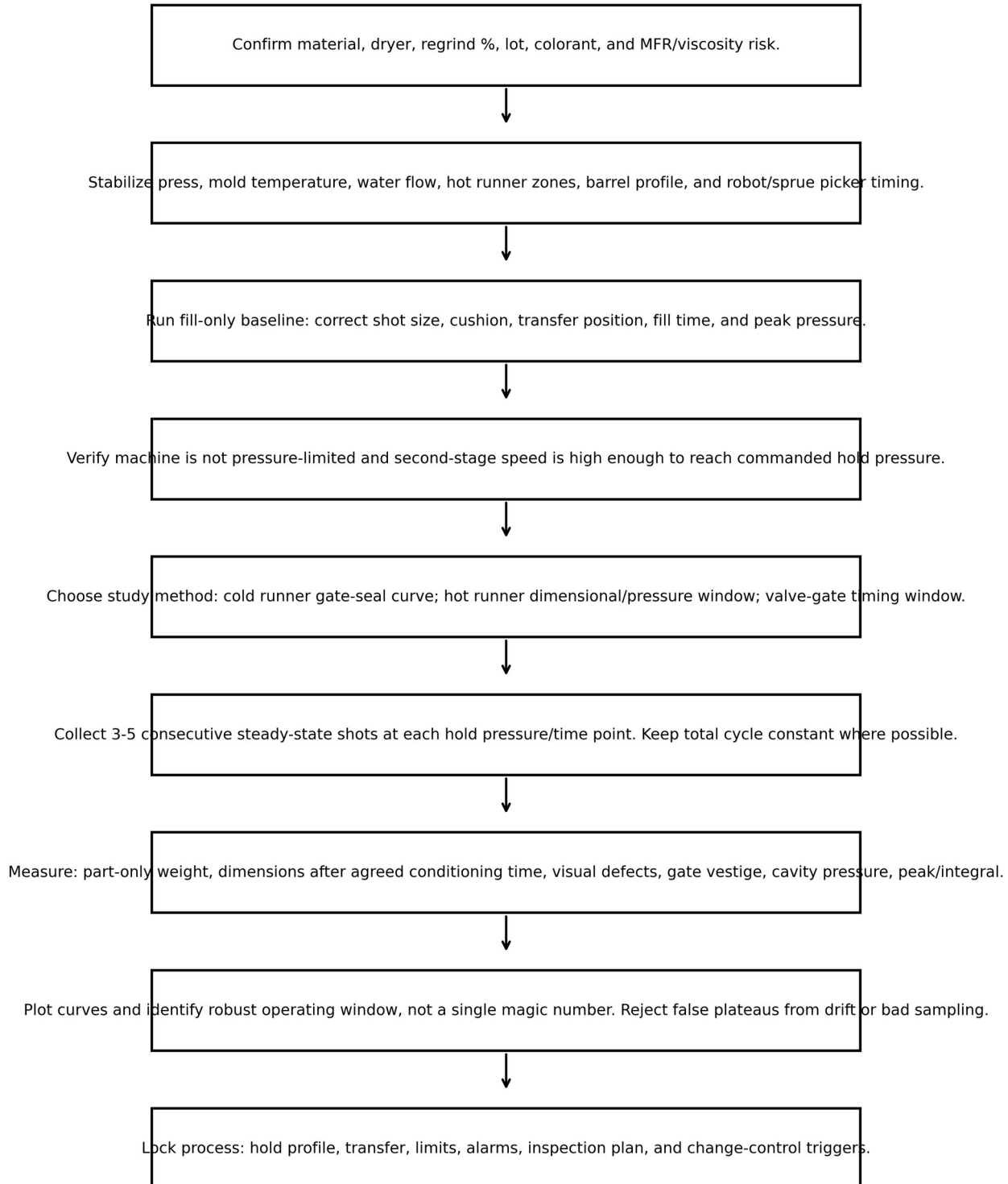
The most realistic near-term advance for shop-floor use is not autonomous magic. It is disciplined data infrastructure: cavity pressure fingerprints, automatic part-weight capture, hot-runner zone history, water-flow

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monitoring, digital setup sheets, machine-state alarms, and statistical comparison between the current curve and the validated reference curve.

10. Troubleshooting Guide and Flowcharts

Hold-pressure study execution flow



10.1 Troubleshooting matrix

Observation during study	Likely cause	Cold runner action	Hot runner / valve gate action	Data to confirm
Weight never plateaus	Large gate, soft olefin, pressure still feeding, runner weight included, hot runner behavior.	Separate part from runner, extend curve carefully, use dimensions/sink, check gate size.	Stop expecting plateau; use cavity pressure/dimensions; check tips/valves.	Part-only weight by cavity, cavity pressure, gate sectioning.
Weight scatter high at same setting	Check-ring leak, unstable melt, scale issue, operator sampling, cushion variation.	Run check-ring test; stabilize material and mold.	Also check hot-runner zone stability and valve actuation.	Cushion trend, actual pressure curve, scale verification.
Sink persists at high hold pressure	Gate frozen too early, thick section isolated, poor cooling, geometry too thick.	Increase gate/relocate gate; redesign rib/boss; improve cooling.	Increase valve-open/pack window if safe; check tip freeze/thermal balance.	Sectioning, CT, pressure at far cavity, simulation.
Flash before sink is fixed	Clamp/mold shutoff/vent limit reached before packing thick area.	Repair shutoffs, reduce pressure, redesign gate/part.	Check valve timing, tips, manifold balance; avoid global pressure increases.	Clamp force calc, parting-line inspection, pressure curve.
Cavity pressure peak varies but machine pressure stable	Material/mold balance changes downstream of screw; gate/tip/vent/cooling variation.	Check runner/gate balance and vents.	Check hot-runner drop temperature and valve pins.	Cavity pressure by cavity, thermal imaging, water flow.
Actual hold pressure does not reach setpoint	Second-stage speed too low, pressure limit, hydraulic issue, screw stall.	Increase allowable second-stage speed within safe range; check machine capacity.	Same, plus check hot-runner restriction.	Machine pressure trace, speed trace, pressure limit alarms.
Dimensions OK at press but drift after 24 h	Post-mold shrinkage/crystallization/stress relaxation, insufficient cooling, moisture conditioning.	Condition parts before final judgement; improve cooling/hold balance.	Same, with hot-runner thermal stability review.	24/48 h dimensions, weight, temperature, material conditioning.
Gate blush or stress cracking	Too much pressure, high shear, small gate, low mold temp, PC/ABS stress sensitivity.	Reduce pressure/time; enlarge gate; increase mold temp if appropriate.	Tune valve/tip temp, reduce pressure spikes, check gate geometry.	Polarized light, visual, pressure overshoot, gate section.

10.2 Common pitfalls that create false determinations

- Running the curve before the mold, hot runner, dryer, and barrel are thermally stable.
- Using total shot weight instead of part-only weight, especially on cold-runner molds with heavy sprues/runners.
- Pooling all cavities together on a multi-cavity mold and hiding cavity imbalance.
- Changing hold time without adjusting cooling time when the objective is to isolate gate seal at constant total cycle.
- Using hydraulic pressure numbers without documenting plastic pressure, screw diameter, intensification ratio, and actual pressure curve.

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- Failing to verify second-stage speed and actual pressure response; the machine may be speed-limited during pack/hold.
- Chasing a weight plateau on a hot-runner or valve-gated mold where the plateau is not physically meaningful.
- Ignoring check-ring leakage, cushion drift, or screw recovery instability.
- Measuring dimensions before parts reach agreed conditioning time, especially semi-crystalline or thick-wall parts.
- Overpressurizing a flashing mold and then treating flash as a normal processing response instead of a tooling/clamp boundary.
- Letting hot-runner zone changes occur during the study without recording them.
- Using one operator's visual judgement as the only quality response.

10.3 Recommended reaction plan

1. Confirm the defect and classify it: underpack, overpack, thermal/cooling, venting, material, machine, hot-runner, or geometry.
2. Return to the validated baseline settings and run five to ten stable cycles. Compare pressure curve, fill time, cushion, part weight, and visual response.
3. If baseline does not reproduce, troubleshoot material, dryer, machine, water, hot runner, and mold condition before changing the process window.
4. If baseline reproduces but quality is unacceptable, rerun the relevant portion of the hold-pressure curve using the current material lot and tool condition.
5. If the curve shows a shifted but still acceptable window, document and approve the adjusted center point through change control.
6. If the curve shows no acceptable window, escalate to tooling/material/design review rather than forcing production with unstable settings.

11. Shop-Floor Templates and Acceptance Criteria

11.1 Hold-pressure curve data sheet

Field	Entry
Mold / part / revision	
Machine / screw diameter / material pressure conversion	
Material grade / lot / color / regrind %	
Dryer settings / moisture result	
Cold runner, hot tip, valve gate, hybrid	
Barrel profile / actual melt temp	
Mold temperature supply-return / flow	
Hot runner zones / valve timing	
Fill speed / transfer position / fill time / peak pressure	
Hold pressure profile	
Hold time values tested	
Cooling time and total cycle control method	
Scale resolution / gage ID	
Dimensions and conditioning time	
Cavity map and sample count	
Approved process window and rationale	
Reviewer signatures	

11.2 Minimum acceptance criteria

Category	Minimum criterion
Repeatability	Fill time, cushion, peak pressure, and part-only weight variation must be stable enough that the curve effect is larger than noise.
Quality response	The selected setting must pass visual, dimensional, functional, and material-specific responses such as stress or void checks.
Process window	The selected setting must have margin to both underpack and overpack boundaries. If no margin exists, do not call the process robust.
Cold runner	Gate-seal/hold-time claim must be based on part-only data, not runner mass. Runner and sprue cooling must be evaluated separately.
Hot runner	No claim of gate freeze based solely on a weight plateau unless the specific gate design and data prove it. Use cavity pressure and dimensions.
Safety	No study point may exceed validated pressure, clamp, mold, hot-runner, or material limits.
Documentation	All settings, actual curves, samples, measurements, and approval decisions must be stored with the process master.

11.3 Peer-review checklist for plant deployment

Use this checklist to convert this reference into a plant-approved procedure. The report itself has not been externally peer-reviewed in this chat; a real shop-floor standard should be signed by qualified personnel who know the plant's machines, molds, materials, and safety systems.

- Process engineer verifies scientific molding sequence, transfer strategy, pressure units, and DOE plan.
- Tooling engineer verifies mold pressure limits, gate design, venting, clamp requirement, and hot-runner/valve-gate constraints.
- Quality engineer verifies sample size, measurement method, conditioning time, acceptance criteria, and traceability.
- Maintenance verifies machine calibration, screw/check-ring condition, water circuits, heaters, thermocouples, and safety systems.
- Production supervisor verifies operator work instructions, sampling discipline, robot/sprue picker setup, and escalation process.
- Material supplier or materials engineer reviews resin-specific drying, shrinkage, degradation, and additive considerations where needed.

12. References and Further Reading

7. **FimmTech**. Scientific Molding: The 6-Step Study, including viscosity, pressure drop, cosmetic process window, and gate seal study. <https://fimmtech.com/knowledgebase-2/scientific-molding-the-6-step-study/>
8. **Plastics Technology / John Bozzelli**. Why and how to do gate-seal experiments; second-stage pack and hold optimization. <https://www.ptonline.com/articles/injection-molding-why-how-to-do-gate-seal-experiments>
9. **Plastics Technology / John Bozzelli**. Gate-freeze study: danger in the name; the test determines hold time needed for repeatable parts. <https://www.ptonline.com/articles/gate-freeze-study-danger-in-a-name>
10. **FimmTech**. Optimizing pack and hold times for hot-runner and valve-gated molds; cold-runner weight plateau versus hot-runner behavior. <https://fimmtech.com/2015/10/26/optimizing-pack-hold-times-for-hot-runner-valve-gated-molds/>

Hold Pressure Curve Study - Injection Molding Technical Reference

11. **Fitzgerald, McDonald, Devine, Fuenmayor.** Transfer and Optimisation of Injection Moulding Manufacture of Medical Devices Using Scientific Moulding Principles, Journal of Manufacturing and Materials Processing, 2021. <https://www.mdpi.com/2504-4494/5/4/113>
12. **Pantani, De Santis, Brucato, Titomanlio.** Analysis of gate freeze-off time in injection molding, Polymer Engineering & Science, 2004. <https://4spepublications.onlinelibrary.wiley.com/doi/abs/10.1002/pen.20000>
13. **Moldex3D.** How to use Moldex3D to assess gate freeze time and optimize packing time. <https://www.moldex3d.com/blog/tips-and-tricks/how-to-use-moldex3d-to-assess-gate-freeze-time-and-optimize-packing-time/>
14. **Moldex3D.** Pack module capabilities: predict gate freeze time, clamp force in packing, packing pressure/time/VP switch, and shrinkage. <https://www.moldex3d.com/products/software/moldex3d/pack/>
15. **Husky Technologies.** Hot Runner Fundamentals; hot runner versus cold runner melt delivery, waste, cycle time, and control differences. <https://www.husky.co/en/resources/blog/hot-runner-fundamentals/>
16. **ARBURG.** GESTICA brochure; ReferencePilot real-time holding-pressure curve control with mold pressure sensor. https://www.arburg.com/media/daten/publications/brochures/ARBURG_GESTICA_523370_en_US.pdf
17. **Kistler.** Cavity-pressure sensors integrated in molds for dynamic pressure-curve measurement and in-process quality monitoring. <https://www.kistler.com/US/en/cavity-pressure/C00000099>
18. **RJG.** Injection pressure in molding; fill, pack, and hold pressure concepts. <https://rjginc.com/injection-pressure-what-is-it-how-to-calculate-it-and-why-it-matters/>
19. **RJG.** Second-stage speed influence on pack/hold pressure transition and speed-limited processes. <https://rjginc.com/how-second-stage-speed-influences-your-process-part-2/>
20. **RJG.** Decoupled III process: fill, velocity-controlled pack to cavity pressure, then hold until gate seal. <https://rjginc.com/tip/when-hold-isnt-hold-in-decoupled-iii/>
21. **Paulson Training.** Scientific injection molding: packing pressure and gate seal determination. <https://paulsontraining.com/scientific-injection-molding-packing-pressure-gate-seal/>
22. **Covestro.** Process variables in injection molding; holding-pressure phase compensates volumetric contraction. <https://solutions.covestro.com/-/media/covestro/solution-center/whitepapers/process-variables---injection-molding.pdf>
23. **SABIC.** Injection molding processing guide; holding pressure generally set high enough to prevent voids/sink marks and control shrinkage. https://www.controlplastics.com/technical-resources/SABIC-PLA-751_Injection_Molding.pdf
24. **Formosa Plastics / Formolene.** PE and PP injection molding process guide; pack/hold pressure commonly 50-75% of injection pressure and hold until gate freeze. <https://www.fpcusa.com/content/uploads/2025/04/Final2025-PE-PP-Injection-Molding-Process-Guide.pdf>
25. **ISO.** ISO 294-1:2017, injection molding of thermoplastic test specimens and reproducible molding conditions. <https://www.iso.org/obp/ui/es/>
26. **ASTM / standards catalog.** ASTM D955 shrinkage measurement from mold cavity to molded dimensions under specified process conditions. <https://standards.iteh.ai/catalog/standards/astm/1674e42b-21dc-4f13-9d8b-3181ccea83f0/astm-d955-08>
27. **SPE Injection Molding Division.** SPE division scope: dissemination of injection molding engineering knowledge. <https://www.4spe.org/Injection-Molding-Division/>
28. **Kistler / Assembly Magazine.** Cavity pressure as key process data for consistent injection-molded parts. <https://www.assemblymag.com/articles/96429-cavity-pressure-is-the-key-to-molding-consistent-quality-parts>
29. **Kariminejad et al..** Ultrasound sensors for process monitoring in injection moulding. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8347947/>
30. **Mohamed et al..** Scanning acoustic microscopy investigation of weld lines in injection molded plastics. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7428748/>

Hold Pressure Curve Study - Injection Molding Technical Reference

31. **ASNT.** Nondestructive testing methods including radiographic and ultrasonic testing.
<https://www.asnt.org/what-is-nondestructive-testing/methods/>
32. **Kariminejad et al..** Single and multi-objective real-time optimisation via Bayesian adaptive DoE in injection moulding. <https://arxiv.org/abs/2402.12077>
33. **Krantz et al..** Machine-learning-based process control for injection molding.
<https://www.osti.gov/pages/biblio/2545121>
34. **Zhao et al..** Recent progress in minimizing warpage and shrinkage deformations by optimization of process parameters. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8831005/>
35. **Mokhtar.** Effect of injection molding process parameters on mechanical properties of ABS and PP polymer, ASEE 2024. <https://peer.asee.org/the-effect-of-injection-molding-process-parameters-on-the-mechanical-properties-of-abs-and-pp-polymer.pdf>
36. **Public practitioner account.** r/InjectionMolding discussion of gate freeze-off and hold-pressure observations; informal, not peer-reviewed. https://www.reddit.com/r/InjectionMolding/comments/1cbbya3/gate_freeze_off/
37. **Eng-Tips practitioner discussion.** Nondestructive testing/inspection for voids in plastic molds; informal, not peer-reviewed. <https://www.eng-tips.com/threads/non-destructive-testing-inspection-for-voids-in-plastic-moulds.53591/>

Final Engineering Position

A hold pressure curve is a high-leverage study because it turns pack/hold from opinion into evidence. On cold-runner molds, the classic gate-seal curve is often appropriate when part-only weight reaches a repeatable plateau. On hot-runner and valve-gated molds, the same method can be wrong; use cavity pressure, cavity-by-cavity weight, dimensions, hot-runner data, valve timing, and a process-window mindset. The safest and most profitable target is not maximum pressure or minimum time. It is the center of a validated window that produces acceptable parts through normal variation without damaging the mold, press, runner system, or product.